Airbus Ship Detection

Traditional machine learning and CNN comparison

Problem

The project classifies satellite images as Ship or No Ship. The target is an image-level label: the system decides whether at least one ship appears in the image. It does not locate ships or produce segmentation masks.

Dataset

The Kaggle run used 192,556 labelled images. The traditional models used 3,000 images and the CNN used a stratified sample of 5,000 images. Images were resized to 128x128. The dataset is imbalanced, with 42,556 ship images and 150,000 no-ship images.

Method

The traditional models use a normalized HSV colour histogram combined with HOG texture and shape features. Random Forest, Linear SVM, KNN, Gaussian Naive Bayes, and Linear Discriminant Analysis were evaluated on a stratified 80/20 split. The CNN loads images in batches and uses three convolution/max-pooling blocks, global average pooling, dropout, and dense layers. Class weights were used during CNN training.

Results

The table below reports the completed Kaggle run. Precision, recall, and F1-score are included because accuracy can be misleading when one class is much larger than the other.

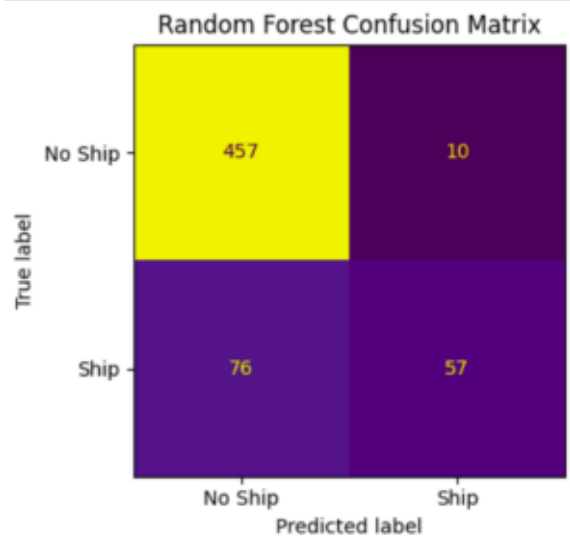
Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.857	0.851	0.429	0.570
Naive Bayes	0.477	0.286	0.910	0.435
Svm	0.713	0.367	0.406	0.386
Lda	0.730	0.347	0.248	0.289
Cnn	0.787	0.562	0.163	0.253
Knn	0.790	0.889	0.060	0.113

Figure files and the full JSON/CSV outputs are included in the repository under outputs/.

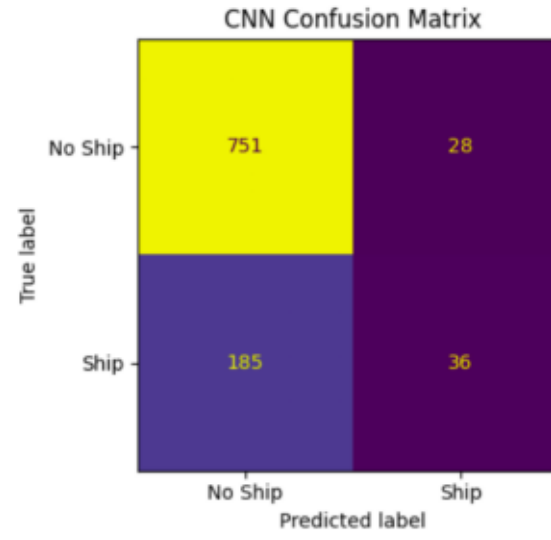
Results and interpretation

The confusion matrices show the trade-off between finding ships and avoiding false alarms.

Random Forest



CNN



Discussion

Random Forest gives the best overall balance in this run, with an F1-score of 0.570 and precision of 0.851. The CNN reaches 0.787 accuracy, but its recall is 0.163 and its F1-score is 0.253, which means that many ship images are still missed at the default 0.5 threshold. Naive Bayes has the highest recall among the traditional alternatives, but its low precision produces many false positives. KNN has high precision but detects very few ships.

Conclusion

The completed notebook provides a reproducible comparison of handcrafted features and a CNN on the Airbus dataset. Random Forest is the strongest baseline in the current experiment. The main limitations are the reduced training samples, the image-level target, and the CNN's low ship recall. Transfer learning, augmentation, threshold tuning, and segmentation-aware modelling are sensible next steps.

Source: completed Kaggle notebook run included with this project.